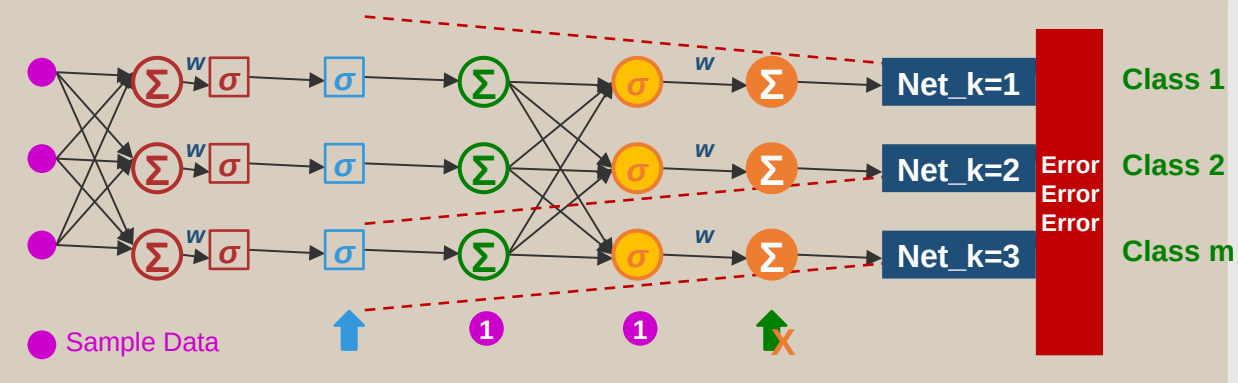
$W_{ji}$ *Affects* $Net_j$ ⇒

$Net_j$ *Affects* $O_j$ ⇒

$O_j$ *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ ⇒

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*




$$= \left[ \frac{\partial E}{\partial \text{Net}_{k=1}} \cdot \frac{\partial \text{Net}_{k=1}}{\partial O_j} + \frac{\partial E}{\partial \text{Net}_{k=2}} \cdot \frac{\partial \text{Net}_{k=2}}{\partial O_j} + \cdots + \frac{\partial E}{\partial \text{Net}_{k=m}} \cdot \frac{\partial \text{Net}_{k=m}}{\partial O_j} \right] \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial Net_k} \Rightarrow -\delta_k$$

$$\text{Net}_k = \sum_k w_{kj} O_j \Rightarrow \frac{\partial \text{Net}_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1 + e^{-Net_j}} \Rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\text{Net}_j = \sum_j w_{ji} O_i \Rightarrow \frac{\partial \text{Net}_j}{\partial w_{ji}} = O_i$$